



Incorporating AI in foreign language education: An investigation into ChatGPT's effect on foreign language learners

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Abstract

ChatGPT, an artificial intelligence application, has emerged as a promising educational tool with a wide range of applications, attracting the attention of researchers and educators. This qualitative case study, chosen for its ability to provide an in-depth exploration of the nuanced effects of AI on the foreign language learning process within its real-world educational context, aimed to utilize ChatGPT in foreign language education, addressing a gap in existing research by offering insights into the potential, benefits, and drawbacks of this innovative approach. The study involved 13 preparatory class students studying at the School of Foreign Languages at a university in Turkey. The students were introduced to ChatGPT through learning experiences over a span of four weeks by the researcher as a language teacher. The qualitative data collected from the interviews were analysed using thematic analysis. The findings suggest that ChatGPT positively affects students' learning experiences, especially in writing, grammar, and vocabulary acquisition, and enhances motivation and engagement through its versatile and accessible nature in various learning activities. These insights contribute to understanding the utility and constraints of employing ChatGPT technology in foreign language instruction and can inform educators and researchers in developing effective teaching strategies and in designing curricula.

Keywords AI in language learning · Artificial intelligence · ChatGPT · Chatbots · Educational technology · Online learning

1 Introduction

Unveiled in November 2022, ChatGPT is an advanced Artificial Intelligence (AI) language model developed by OpenAI (OpenAI, 2024). It is built upon the foundation of the Generative Pre-trained Transformer-3 (GPT-3) (Korngiebel & Mooney, 2021). As a large language model (LLM), it boasts a staggering 175 billion parameters

(Cotton et al., 2023; Topsakal & Topsakal, 2022; Wenzlaff & Spaeth, 2022; Zhai, 2022), enabling the generation of text responses akin to human-composed writing (Shen et al., 2023). Leveraging deep learning techniques like deep neural networks and deep belief networks, as well as cortical algorithms that outperform deep belief networks, ChatGPT uses advanced architectures and algorithms to efficiently train on large datasets, enabling complex hierarchical representations and transformations that power its capabilities across voice recognition, chatbots, search, and computer vision (Moshayedi et al., 2022; Rizk et al., 2019; Shrestha & Mahmood, 2019). Owing to these capacities, ChatGPT can exhibit a plethora of natural language-processing capabilities, which include resolving queries, spinning narratives, conducting logical assessments, debugging programming codes, and facilitating machine translation (Ouyang et al., 2022). Functioning as a general-purpose conversational agent, ChatGPT and other large language models have the potential to be indispensable tech-tools in various fields, including but not limited to education, healthcare, software engineering, content creation, translation, business, cost reduction, and customer service (Aljanabi, 2023; Brown et al., 2020; Gunawan, 2023).

The existing literature unveils an ever-growing curiosity about the influence of AI, notably ChatGPT, on language acquisition. The integration of these technologies into language teaching exhibits varied strategies and perceptions, indicating both benefits and limitations, especially in processing certain languages (Bao & Li, 2023; Moorhouse, 2024) and challenges in addressing deeper linguistic issues (Algaraady & Mahyoob, 2023).

Teacher and student perceptions of AI in language-learning environments present a nuanced picture. AI's potential to enhance educational experience, as seen in studies by Guo and Wang (2023) and de Vicente Yagüe Jara et al. (2023), which emphasizes AI's role in providing comprehensive feedback and fostering language skills, and varied opinions among faculty on its effectiveness (Mohamed, 2023) suggest a landscape of cautious optimism tempered by skepticism. Sociocultural factors significantly influence learners' attitudes toward AI, as observed in studies of Korean middle school students (Kim & Lee, 2023), emphasizing the importance of contextual considerations in the adoption and integration of AI in educational settings.

The integration of AI into curriculum design and teaching methods suggests potential for innovation. Proposals for employing natural language processing and peer feedback within digital environments (Bauer et al., 2023) and the exploration of student-AI collaboration (Kim et al., 2022) indicate forward-thinking approaches to curriculum development. These strategies underscore the need for AI education and supportive environments to maximize the benefits of AI in language learning. In terms of language assessment and feedback, AI shows promise for enhancing core language skills (Ayotunde et al., 2023). Lee et al. (2023) highlighted the effectiveness of combining ChatGPT with prompt engineering in automatic question generation, suggesting a new avenue for creating educational content in English learning. However, limitations in identifying deep structural and pragmatic errors in learners' writing (Algaraady & Mahyoob, 2023) call for a balanced approach that leverages AI capabilities, while recognizing the indispensable role of human expertise. Yang et al. (2023) delved into the dynamics of AI-generated feedback interaction, noting that Chinese EFL students prioritize error correction, a finding that complements discussions on feedback mechanisms in language learning. Similarly, Zaki et al.

(2023) introduced an AI-based system for mapping learning outcomes, reinforcing AI's utility beyond teaching and administrative tasks such as quality assurance in education.

Challenges and ethical considerations are significant to the discourse on AI in education. Concerns regarding academic integrity, the pedagogical implications of AI-generated texts (Barrot, 2023; Rawas, 2023), and the difficulties faced by ESL instructors in identifying AI-generated content (Alexander et al., 2023) highlight the complexities of incorporating AI into language learning frameworks. Furthermore, the potential impact of AI on job devaluation among language service providers necessitates a strategic approach for its adoption (Tavares et al., 2023).

Finally, the potential and limitations of AI, including generative tools such as ChatGPT, in enhancing language skills are evident. While such technologies offer valuable research applications and can improve writing quality (Marzuki et al., 2023), ethical concerns and the balance between innovation and responsible use remain critical considerations (Pack & Maloney, 2023). The literature thus delineates a landscape where AI's contributions to language learning are significant, yet circumscribed by challenges that require careful navigation to ensure ethical, effective, and inclusive educational practices.

This study narrows down its lens to examine the employment of ChatGPT in the domain of foreign language learning. In doing so, it aims to fill an existing research gap, offering new insights regarding the possibilities, advantages, and challenges tied to ChatGPT. Tlili et al. (2023) states that while the use of ChatGPT within the educational field sparks debate, its proficiency in creating summaries, quizzes, and flashcards could bolster learning experiences (Khan et al., 2023). While ChatGPT can generate seemingly accurate and coherent text, concerns remain about the reliability and ethical use of its output, especially in academic contexts where it may enable plagiarism and undermine learning. (AlAfnan et al., 2023; Kung et al., 2023; Perkins, 2023). Therefore, this study investigated the effects of ChatGPT on students' language learning experiences as well as their views on the various benefits and limitations of ChatGPT in foreign language education. Our study also explored the positive and negative aspects of ChatGPT's implementation in the language class. By examining the students' views on both the advantages and drawbacks of ChatGPT's utilisation, our study provided an in-depth knowledge on the applicability and effectiveness of AI use in foreign language education. To comprehensively investigate the role of ChatGPT in foreign language learning, the study was guided by the following research questions:

1. How does the implementation of ChatGPT in foreign language learning affect students' learning experiences?

This question aims to explore the impact of ChatGPT on various aspects of the foreign language learning process, including students' engagement, motivation, and skill acquisition.

2. What strengths, challenges, and recommendations emerge from student perspectives on the use of ChatGPT for foreign language learning?

Through this question, the study seeks to gather insights on the perceived advantages and limitations of ChatGPT from the learners' viewpoint, as well as suggestions for its effective integration into language education.

Ultimately, the findings from this study will illuminate the benefits and drawbacks of ChatGPT in foreign language instruction, guiding educators and researchers in the development of effective language teaching strategies.

2 Methodology

2.1 Research design

A qualitative case study design was used in this study. According to Yin (2009), a case study design is preferred for several reasons, such as: (1) conducting an in-depth examination of a situation within its actual context and temporal setting; (2) utilizing it to gain insights into new approaches, theories, or models for understanding cases. Cases are treated as a unit of analysis in the research, with the design process determining what constitutes a case (Patton, 2002). In this study, “students from different programs and with different language levels” were considered cases. In this context, the researchers examined ChatGPT’s effects on students’ language learning experiences. Additionally, they explored the tool’s advantages and drawbacks in foreign language education. Figure 1 illustrates the overall design and flow of the research.

2.2 Study group

The study group consisted of 13 preparatory class students studying at the School of Foreign Languages at a foundation university in Ankara, Turkey. In the school of foreign languages, where the research was conducted, students take exams (formative assessments) every two months. This study used the scores from those regular exams instead of creating new ones. To this aim, a purposive sampling method—specifically, maximum variation sampling—was employed. According to Patton (2002), maximum variation sampling aims to reach common judgements and experiences starting from diversity. The reasoning is that students from various departments would enrich and diversify the data of the study. In addition, the diversity of the participants in terms of language proficiency was clearly emphasised by representing participants from all levels in the A2-B2 range in the sample. Since ChatGPT is an emerging technology, administrators at the university where the research was conducted were concerned about its implementation in classes. Consequently, they allowed researchers to use it only in a single classroom. Therefore, the research was conducted with volunteers from one researcher’s class. The reason for the online class was the state of emergency distance education after a massive earthquake hit Turkey on 6 February 2023. The demographic characteristics of the participants are presented in Table 1.

As shown in Table 1, the study group of the research included 10 female and 3 male students. All of the participants were aged between 18 and 20. Students from different departments, ranging from gastronomy to dentistry, constituted the study group. The students’ language levels ranged from A2 to B2 as described in

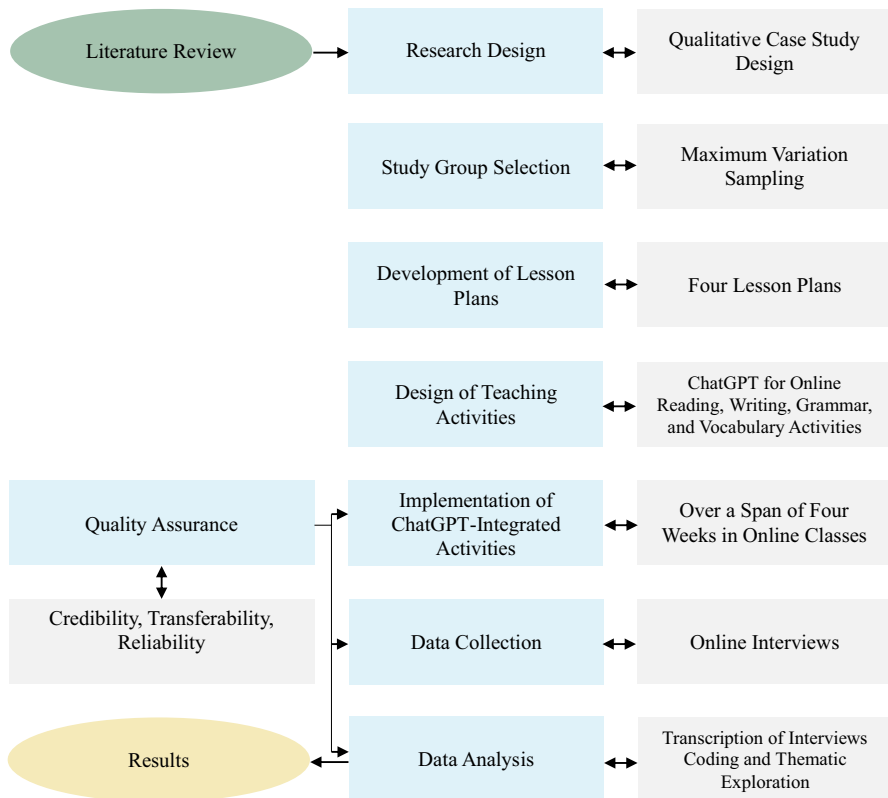


Fig. 1 Research design and flow

the Common European Framework of Reference for Languages. Additionally, while the personal information form did not include a question concerning the students' previous experiences with using ChatGPT for learning a foreign language, most students reported using the tool for the first time for this purpose during face-to-face interviews.

2.3 Program description

The program aimed to actively engage students in online classes using ChatGPT while also guiding them in how to use the tool effectively for foreign language learning. ChatGPT was integrated into online reading, writing, grammar, and vocabulary classes, each lasting 45 min. Four lesson plans were tailored based on the four units of *Reading Explorer 2* by *National Geographic Publishing*, which was used as the coursebook in Reading and Writing classes in the Foreign Languages Department of the university where the study was conducted. Additionally, teaching activities were designed by the researchers to empower students in enhancing their writing and grammar skills through the effective use of ChatGPT.

Table 1 Demographic characteristics of the participants

Learner code	Gender	Age	Department	Language level
L1	Female	19	Political Science and International Relations	A2-B1
L2	Female	18	International Trade and Financing	B2
L3	Female	18	Gastronomy and Culinary Arts	B1
L4	Female	19	Electrical and Electronics Engineering	B2
L5	Male	20	Department of Electrical and Electronics	B1
L6	Male	20	Computer Engineering	B2
L7	Female	20	Nursing	B1
L8	Male	19	Computer Engineering	B1
L9	Female	20	Dentistry	A2
L10	Female	19	Gastronomy and Culinary Arts	B1
L11	Female	20	Political Science and International Relations	A2
L12	Female	18	Political Science and International Relations	B2
L13	Female	19	Gastronomy and Culinary Arts	A2

2.4 Teaching materials

During the four-week research period, the teaching materials included tailored lesson plans, slides, and ChatGPT. For the Reading and Writing lessons, the lesson plans were composed of the themes of “*City Life*, *Backyard Discoveries*, *When Dinosaurs Ruled*, and *The Brothers Grimm*”. The students were encouraged to exploit ChatGPT to create ideas, make stories, or write sentences related to the reading topics or newly learned vocabulary. In writing classes, students were taught how to write a ChatGPT-assisted classification and a cause-and-effect essay through brainstorming ideas, making plans, and receiving written feedback. Finally, in ChatGPT-assisted grammar classes, students had the opportunity to practise past tenses, modals of deduction, reported speech, and noun clauses for fun grammar practice and for the consolidation of grammar presentations.

2.5 Language learning activities

The four-week ChatGPT-assisted English course encouraged students to effectively use the tool in online English lessons. The reading and writing lesson plans comprised of ChatGPT-assisted warm-up, pre-reading, reading comprehension, vocabulary practice, and wrap-up activities. ChatGPT was integrated into four lesson plans, and Fig. 2 shows the various ways students engaged with ChatGPT in different language learning activities.

As Fig. 2 illustrates, students engaged with ChatGPT in various language learning activities. For example, in the first week, students used it to create social media posts for the “global cities” theme and to generate hashtags and captions. Figure 3 depicts the social media posts and hashtags two participants created for London and Brussels.

Social Media Simulation	Group work	Students used ChatGPT to create social media posts about global cities.
Story Generation	Group work	Students used ChatGPT to generate short stories.
Virtual Field Trip Planning	Individual and Group Work	Students planned virtual field trips to natural places using ChatGPT.
Role-Play Creation	Group Work	Students created role-plays using specific vocabulary words and presented them to the class.
Research Proposal	Group Work	Students used ChatGPT to brainstorm research questions and generate project proposals
Vocabulary Practice	Individual and Group Work	Students used ChatGPT to create posters and quiz questions to practice specific vocabulary
Pre-reading Activity	Group Work	Students explored topics or information related to the reading text with the help of ChatGPT.
Wrap-Up Activity	Individual and Group Work	Students engaged in discussions, debates, or conversations facilitated by ChatGPT.
Writing Feedback	Individual Work	Students used ChatGPT to receive feedback on their essays.
Grammar Practice	Individual and Group Work	Students practiced specific grammar concepts, such as past tenses, modals, and noun clauses.

Fig. 2 ChatGPT integration in language learning activities

Similar to the activity presented in Fig. 3, the students also used the tool to generate answers for questions related to target vocabulary to discuss with peers in breakout rooms. Similarly, as a wrap-up activity, the students worked in groups of two on a ChatGPT-assisted creative writing exercise. Each student in the group was given a prompt by the teacher, which they pasted into the ChatGPT to brainstorm, plan, and write a story. The students took notes on the story provided by ChatGPT and shared their stories with their partners in the breakout rooms in the online class. The students were encouraged to ask questions and engage in discussion about each other’s stories. In terms of prompts, the teacher provided students with prompts in the first week. However, in the following weeks, the students were encouraged to write their own prompts to foster independence and creativity. Moreover, regarding the use of

"Feeling the hustle and bustle of London today! From the iconic red telephone booths to the towering Big Ben, there's never a dull moment in this amazing city. Can't wait to explore more! 🇬🇧 #LondonCalling #CityOfDreams #TravelGoals #UKadventures"

Caption: Fall in love with the charming city of Brussels 🇧🇪 From delicious chocolate and waffles to stunning architecture and art, there's something for everyone in this beautiful city.

Hashtags: #Brussels #Belgium #Travel #Europe #Chocolate #Waffles #Architecture #Art #CityBreak

Fig. 3 Social posts and hashtags created with ChatGPT

ChatGPT, students were encouraged to use ChatGPT's answers while also being guided to include original and responsible ideas in their responses.

2.6 Data collection and analysis process

The study conducted to examine the effectiveness of ChatGPT-assisted language learning on students' learning experiences was granted an ethical approval by the Ethics Committee of the university where the research was conducted. Following this, the course instructor clearly explained the study's purpose and data confidentiality to the students and obtained their consent. Thereafter, the students in the study group were subjected to a four-week series of activities implemented by the course instructor.

An abrupt shift from in-person classes to emergency distance education was mandated by the Council of Higher Education due to a devastating earthquake in Turkey. Therefore, this study and interviews were conducted online. Students participated in online classes using Microsoft Teams, while interview sessions were conducted using Zoom.

To schedule the interviews, participants were initially asked to fill out a Google Form to indicate their preferences. The interviews were then conducted in accordance with the time slots selected by the participants. Participants were provided with an overview of the research's objectives, and their consent was obtained for their voluntary participation in the interview process, including video recording. Each interview lasted about 15–20 min, and there were no issues with video, audio, or presentation screens during the interviews.

The semi-structured interview form questions were developed and subsequently analysed within the framework of Yin's (2011) understanding of case studies. Accordingly, (1) semi-structured interview questions were formulated in accordance with the research questions. Within the framework of the research questions, two review articles (Zawacki-Richter et al., 2019; Yang & Kyun, 2022) that provide a broad perspective on artificial intelligence applications in foreign language learning were examined in detail in the formulation of semi-structured interview questions. Based on these studies, interview questions were developed around the themes of 'ChatGPT in Foreign Language Learning' encompassing its applications, impacts, benefits, drawbacks, and optimization in language learning (see Appendix 1). Moreover, they were asked for suggestions and recommendations on how to use ChatGPT more effectively in language classrooms. After the interviews, (2) the audio recorded interviews were transcribed in written forms, (3) multiple researchers were included in the analysis to establish a coding

and category system suitable for the research questions (e.g., code: Limited aspects of ChatGPT. Category: Encourages taking the easy way out by providing information ready-made), (4) each researcher independently assigned the codes obtained from the transcription to the code-category system, (5) similarities and differences among the assigned codes were examined, (6) to ensure agreement between the two researchers doing the analysis, their coding systems were combined and any differences were reviewed and analyzed on a Microsoft Word file until they reached a common understanding, and (7) findings were reported.

2.7 Validity and reliability

Validity illustrates the extent to which research data is consistent with real life and is examined in two main categories: internal validity and external validity (Lucas, 2003). In qualitative research, credibility is the favored term for internal validity, while transferability is used for external validity. Credibility measures if the research findings are accurate and correct. Contrarily, external validity, or transferability, concerns the extent to which the research findings can be generalised to diverse contexts (Merriam, 2009).

To corroborate the credibility of the research, the preparatory class students forming the study group were introduced to ChatGPT through activities conducted over a span of four weeks by a researcher adept in the field of language education. Following this, four researchers with experience in language education prepared the semi-structured interview form based on experiences and feedback derived from the conducted training. The questions were reviewed for clarity and language by a panel of five experts, comprising three language educators and two specialists in education, resulting in necessary amendments.

During the preliminary application, post-interview observations confirmed no issues in understanding the research questions. Since no modifications were required for the questions, the data procured from the preliminary application interview was incorporated into the final application. After engaging in a prolonged interaction with ChatGPT, the students expressed their experiences and views via the interview form. Additionally, the researcher's extended presence in the field facilitated critical interpretation from the participants' perspectives while presenting the findings, thereby revealing patterns in a comprehensive manner.

The transferability of the research was secured through the application of the maximum variation sampling method, a purposeful sampling technique. Consequently, an examination of preparatory class students from various departments' experiences regarding ChatGPT was conducted, intending to reflect diversity in the data by presenting disparate perspectives. The codes and themes are clearly exhibited to aid researchers working on similar topics. Moreover, the findings are meticulously depicted with direct quotations derived from the interview data.

In qualitative research, reliability refers to the consistency between the phenomena occurring in the real environment and the data researchers have acquired: in other words, the consistency between the accuracy and comprehensiveness of the data (Bogdan & Biklen, 2006). The data should be context-specific and articulated in a detailed and in-depth manner (Cohen et al., 2017). In this research, interviews were conducted immediately after the ChatGPT training sessions with participants

to minimise the chances of their thoughts regarding ChatGPT deviating from the context and situation. Accordingly, researchers adopted a similar approach, posing identical questions to each participant. Moreover, all interviews were recorded for future verification of the data. Measures were taken to ensure students' participation throughout the activities and interviews, and potential impacts from external factors were minimised through appropriate control measures. To enhance the internal reliability of the qualitative data collection process, the data was evaluated through inter-coder reliability (ICR). For this purpose, the data was independently coded by two researchers.

In this research, two researchers, who received their doctoral theses in curriculum & instruction and specialized in foreign language education, demonstrated their proficiency in qualitative data analysis methods and data coding and interpretation. Leveraging their deep expertise in qualitative data analysis, they thoroughly interpreted the collected data. Utilizing the 'simple percentage agreement' formula by Miles et al. (2019), they achieved an impressive 86% consensus level. The researchers engaged in a meticulous thematic analysis for Tables 2 and 3, independently coding interview responses to unearth emergent patterns and commonalities. For instance, the "Impact on specific language skills" section in Table 2 highlighted varied student perceptions, such as one student (L3) emphasizing ChatGPT's role in contextual vocabulary learning and another (L5) appreciating the immediate feedback on word usage. Table 3's categories, encompassing students' views on strengths, challenges, and recommendations for language learning, were directly shaped by the research question to maintain relevance and coherence. This methodological rigor ensured a robust, grounded interpretation of the qualitative data, aligning closely with the students' perspectives and the research inquiry.

3 Results

The study investigated the effectiveness of ChatGPT-assisted language learning on students' learning experiences and explored their views about its benefits and limitations in foreign language learning. The study also explored the positive and negative aspects of ChatGPT's implementation. Thirteen students were interviewed. Anonymity was ensured by using code names for the students (for example, L1, L2, L3.etc.), which are consistently used across all tables. Students' views on ChatGPT were analysed and organised into three main themes and presented under these themes with detailed subheadings.

3.1 The impact of ChatGPT on students' foreign language learning experiences

A qualitative analysis was conducted to answer the first research question. Patterns of the students' views on the impact of ChatGPT implementation on students' learning experiences have been identified and presented in Table 2.

Table 2 ChatGPT's impact on students' foreign language learning experiences

Themes	Category
Impact on specific language skills	Writing improvement: L5, L6, L8, L12 Grammar enhancement: L3, L4, L5, L10, L11, L12 Vocabulary learning: L1, L8, L12 Limited impact on speaking: L4, L12 No impact on listening: L8
Motivation and engagement	Enhanced motivation: L1, L7, L8, L9, L10, L12 Engaging learning experience: L8, L9 No increase in motivation: L4, L6
Versatility in learning activities	Creative writing activities: L3, L5 Interactive learning experiences: L2 Cultural exploration opportunities: L9
Areas for improvement	Occasional technical issues: L7 Limited contribution to language learning: L4

The findings suggest that ChatGPT-assisted language learning has a positive impact on certain language skills, particularly on writing and grammar skills, as well as on new vocabulary acquisition (L1, L3, L4, L5, L6, L8, L10, L11, L12). One student (L3) highlighted this by stating, “*ChatGPT helped me learn new words in context, making it easier to remember them.*” However, it may have limited impact on speaking skills (L4, L12) and no impact on listening skills (L8). According to the participants, ChatGPT improves students' motivation and learning engagement, contributing to a more engaging learning experience (L1, L7, L8, L9, L10, L12). For instance, a student (L7) appreciated the tool for its instant feedback, mentioning, “*It corrected my word usage instantly, which was very helpful.*” Additionally, the participants emphasised ChatGPT's adaptability for various learning experiences, such as interactive learning, creative writing, and exploring different cultures, as well as its ease of access and quick feedback (L3, L2, L5, L9, L10). However, a few students mentioned technical difficulties faced with ChatGPT and finally its limited assistance in making language learning easier, suggesting these as potential areas for enhancement (L4, L7). It can therefore be assumed that ChatGPT carries the potential to enhance language competency while fostering motivation and engagement amongst learners. However, it remains essential to continue refining the system to mitigate its constraints and enhance the language acquisition process.

3.2 Student views on using ChatGPT for foreign language learning: Strengths, challenges, and suggestions

Diving into the second point of our inquiry, we delved into student perspectives on the strengths, challenges, and suggestions of ChatGPT as a tool in the language learning process, in addition to their propositions on how its performance could be upgraded. The key points of these conversations can be found in Table 3. Table 3

Table 3 Strengths, challenges, and recommendations on ChatGPT for foreign language learning

Themes	Category
Strengths	Writing skills improvement (L1, L4, L9, L11)
	Time-saving and efficient research (L1, L3, L4, L13)
	Comprehensive knowledge and assistance (L1, L2, L10)
	Cultural and creative activities (L1, L2, L10)
	Vocabulary and sentence expansion (L1, L5, L8, L9, L11)
	Grammar learning and examples (L3, L4, L6, L7, L8)
	Games and Flexible use (L1, L5)
Challenges	Dependency and skill deterioration (L1, L2)
	Connection issues and errors (L1, L3)
	Encouragement of shortcuts (L1, L2, L3)
	Limited effectiveness for speaking practice (L11)
	Misuse, Emotional Responsiveness, and Accessibility Issues (L3, L5)
Recommendations for improvement	Technical enhancements and increased accessibility (L1, L4, L5)
	Integration with real-life situations and interactivity (L2, L5)
	Novelty and engagement (L2, L5)
	Efficiency and time-saving (L3, L13)
	Addressing potential misuse and limitations (L2, L3)
	Improved communication and personalization (L3, L4)

illustrates the numerous perspectives from which ChatGPT can be regarded in relation to foreign language learning.

Students' perspectives underscored the positive impacts of ChatGPT in foreign language learning. The students praised how ChatGPT enhanced their writing skills, as expressed by one student (L9): *"I realized that I can express myself better in writing thanks to ChatGPT."* They also appreciated how ChatGPT expedited research processes by being time-saving and efficient (L1, L3, L4, L13), with a student (L1) noting, *"I can access the information I need faster."* The tool also provided comprehensive knowledge (L1, L2, L10) and contributed to cultural and creative activities (L1, L2, L10), with one student (L10) observing, *"It even gives me cultural information about the words."* Additionally, they mentioned the improvement of vocabulary and sentence structures (L2, L5, L8, L9, L11) and the enhancement of grammar skills through practical examples (L3, L4, L6, L7, L8). Some students found the tool useful for games or flexible uses (L1, L5).

Despite these positive aspects, students also identified some challenges and potential drawbacks of using ChatGPT. There were concerns about possible over-reliance on ChatGPT (L1, L2), with a student (L1) expressing, *"I am afraid I will get used to it too much and my English will deteriorate."*

Other concerns included occasional connection issues and errors (L1, L3), encouragement of shortcuts in learning (L1, L2, L3), and limited effectiveness for speaking practice (L11). Students also highlighted potential misuse during assignments or exams (L3) and called for improved responses to emotions and personal

expressions (L3). Some students pointed out issues of inaccessibility due to high traffic (L5).

Students also offered several recommendations for enhancing the effectiveness of ChatGPT in language learning. They called for technical enhancements and increased accessibility (L1, L4, L5) and integration with real-life situations and interactive learning experiences (L2, L5). The students saw value in the tool's novelty and engagement (L2, L5) and urged for its better efficiency and time-saving capabilities (L3, L13). Furthermore, they called attention to potential misuse and limitations (L2, L3) and suggested improved communication and personalization, such as more natural and human-like responses (L3, L4).

4 Discussion

4.1 The impact of ChatGPT on students' foreign language learning experiences

The result indicates that ChatGPT positively impacts language skills, particularly in writing, grammar, and vocabulary acquisition, and increases motivation and engagement among students for its versatility and accessibility in various learning activities. Studies on ChatGPT-assisted language learning, as well as systematic reviews on the effectiveness of AI-powered language learning tools, confirm the significance of these technologies in enhancing linguistic competencies and learner engagement (Shadie & Liang, 2023). They reported learners experiencing positive learning outcomes, increased motivation, engagement, and confidence, despite some identified challenges. Such advancements are crucial in language acquisition, providing a foundation for more advanced linguistic competencies. In another study, Song and Song (2023) found that AI-assisted instruction using ChatGPT led to significant improvements in EFL students' writing skills, motivation, organization, coherence, grammar, and vocabulary compared to traditional instruction. Similarly, Nugroho et al. (2023) highlighted that ChatGPT could stimulate authentic interactions and boost student productivity and engagement in language learning activities. Rawas (2023) also highlighted ChatGPT's potential to enhance personalization and collaboration in higher education, though ethical and implementation challenges exist. Additionally, Klayklung et al. (2023) emphasized ChatGPT's ability to provide personalized learning experiences and facilitate meaningful interactions, highlighting its potential in language learning. These perspectives complement our observation of increased motivation and engagement among students for ChatGPT's accessibility in learning activities. The findings of the current study are also consistent with previous research (Ali et al., 2023; Habeb Al-Obaydi et al., 2023; Kim et al., 2023; Kohnke et al., 2023; Leonard et al., 2023; Niyozov et al., 2023; Tipayavaravan et al., 2023; Yan, 2023) which support the effectiveness of ChatGPT in language learning and its potential to create an enhanced language learning experience and to revolutionize language pedagogy, although further investigation is needed to fully understand its impact and potential in various learning environments. Guo and Wang (2023) observed that ChatGPT offers more comprehensive feedback in EFL writing than human teachers, albeit with mixed teacher perceptions. Further supporting this, Hwang et al. (2023) demonstrated significant improvements in EFL

writing through AI-Generated Sample Sentences (AI-SS) and AI-Writing Feedback (AI-WF), which resonates with our findings on ChatGPT's role in improving writing skills. The interactive and feedback-oriented nature of ChatGPT evidently aligns with these emerging pedagogical tools, suggesting a broader trend in the effective use of AI in language education. Furthermore, Topsakal and Topsakal (2022) emphasized the potential of ChatGPT in creating interactive dialogs and content, suggesting its applicability in language learning environments. Although ChatGPT provides an interactive learning experience, it lacks the benefits of communicative practice, such as spoken conversation in a cultural context and emotional support. McCallum (2023) discussed the use of ChatGPT in designing culture-focused telecollaboration tasks, highlighting the tool's potential in developing intercultural communicative competence. This aspect is particularly relevant in the context of foreign language learning, where cultural understanding plays a key role.

4.2 Students' perspectives on the advantages and disadvantages of ChatGPT in foreign language learning

The results highlight some advantages of ChatGPT-assisted language learning. Firstly, ChatGPT could improve the participants' English writing skills by providing them ideas, feedback, and introducing them to different grammatical structures and wordlists. This finding is supported by the previous studies emphasising the contribution of AI language models on vocabulary skills development (Alsadoon, 2021) and writing feedback provision (Dai et al., 2023). Akopiants (2023) emphasizes the potential of ChatGPT in enhancing vocabulary skills in English language learners. These results underscore the potential of ChatGPT in augmenting traditional language learning methods, particularly in writing skill development. Rospigliosi (2023) also concluded that ChatGPT fosters learner-centered teaching and enhances creativity, problem-solving abilities, and vocabulary development in students. Additionally, Song and Song (2023) revealed significant improvements in various aspects of writing, including vocabulary, among students who received AI-assisted instruction compared to those with traditional instruction. As a second result, ChatGPT was reported to ease the research by saving the students' time. The previous studies support this finding as ChatGPT-assisted language learning served as an efficient tool to provide useful information and enables the students to access information effectively, to generate lesson plans and save time by creating teaching materials, worksheets, and assessments (Farrokhnia et al., 2023; Fauzi et al., 2023; Firat, 2023; Stepanchko & Kozub, 2023). Such efficiency not only streamlines the learning process but also encourages students to engage more deeply with the language learning content. Thirdly, ChatGPT-assisted language learning was reported to provide an opportunity to engage in cultural and creative learning experiences. Michalon and Camacho-Zúñiga (2023) and McCallum (2023) suggest that ChatGPT can significantly aid in developing critical thinking and communication skills, as well as intercultural competencies, which are vital in language learning. This supports our findings on ChatGPT's effectiveness in improving writing skills and fostering creative and cultural learning experiences. The participants mentioned that ChatGPT-assisted language

learning helped them to be more creative by generating different and creative story endings. This result is well-supported by the findings of Zhai (2022), who mentions that ChatGPT is essential to improving creativity in teaching and learning process.

Regarding the disadvantages of ChatGPT-assisted language learning, the first major concern raised by participants was the possible over-reliance on the tool. This apprehension about becoming too dependent on ChatGPT aligns with concerns documented in existing literature. Graves (2023) cautions about the potential risks associated with the normative standards of language and performance in AI-assisted learning environments. Emphasizing the importance of maintaining strong teacher-student relationships and advocating for the use of AI as a supplement to traditional teaching methods, Mohamed (2023) recommends that teachers should continue to play a vital role in the language learning process and exercise caution in their integration of AI technology. This is to avoid the risk of students becoming overly reliant on AI systems and to ensure that students benefit from a balanced approach that incorporates both technology and human interaction. This concern is further echoed by Sampson and Yoshida (2020), who highlight the significance of considering the emotional and psychological aspects of language learning facilitated by technology. Additionally, Sun and Hoelscher's study (2023) mentions that while helping learners develop reflective and critical thinking, this situation may impede the development of critical thinking skills and learner autonomy. Mohammad et al. (2023) also emphasized ChatGPT's, generation of incorrect information, biases, potential to undermine critical thinking skills, ethical concerns such as cheating, and privacy issues. This is also supported by Vaccino-Salvadore (2023) who discussed the ethical dimensions of using ChatGPT in language learning, including privacy, bias, reliability, accessibility, authenticity, and academic integrity concerns. Therefore, the students should be informed to be cautious against the drawbacks of ChatGPT in the learning process. Finally, the students indicated a few occasional technical problems such as connection issues. This finding is supported by the previous studies highlighting technical drawbacks of using chatbots (Crawford et al., 2023; Huang et al., 2022). These insights suggest that while ChatGPT and similar AI tools offer significant benefits, their role should be viewed as complementary to traditional teaching methods, ensuring a balanced and holistic approach to language education.

4.3 Limitations and recommendations for future research

One of the most significant limitations of the study is the reliance on a limited and homogeneous group of 13 students. This may mean that the findings are not generalizable across different age groups, cultures, and educational levels. The results of the study may not comprehensively represent a broader student population and various learning styles. In this context, future research should be conducted with larger and more diverse sample groups. Additionally, technological access and internet connectivity issues may have prevented students from utilizing ChatGPT to its full capacity. The fact that students used the free version of ChatGPT (ChatGPT 3.5) also raises questions about whether the performance and impact on foreign language learning might differ with the premium version (ChatGPT 4). At this point,

as Mohamed (2023) also recommends, guidelines and protocols for the ethical and responsible use of ChatGPT in EFL teaching and learning should be developed. The qualitative data reflecting participants' personal opinions could complicate objective inferences; therefore, future research involving more extended quantitative studies to investigate the effectiveness of ChatGPT in influencing learners' academic success, motivation, and belief in foreign language learning and attitudes toward the use of artificial intelligence in foreign language education would support participant bias in a more valid, reliable, and generalizable manner.

4.4 Pedagogical implications

The outcomes of this study hold significant pedagogical implications for EFL students. (i) ChatGPT presents language learners with the opportunity to personalize and manage their learning processes, allowing them to follow their own pace and styles. This affords students the chance to explore and craft their own language learning journeys. (ii) Furthermore, the interactive nature of ChatGPT fosters more active and motivated engagement in the language learning process. By offering opportunities for writing, reading, and grammar practice, ChatGPT aids in enhancing students' language skills in various ways while also providing different perspectives on creative writing and language usage, thus making the language learning process more creative and interactive. Aligning with these findings, the research conducted by Karataş (2023) revealed that participants recognized the significant benefits of ChatGPT, particularly in its role in facilitating the creation of learning materials, engaging students through interactive questioning, and overall enrichment of the teaching process. However, it is observed that ChatGPT has not yet been effective in improving speaking skills, whereas other AI-based language learning applications like "Elsa Speak" have been shown to be beneficial in developing speaking skills, particularly pronunciation ability (Akhmad & Munawir, 2022). (iii) On the other hand, thanks to the variety of content and learning materials provided by ChatGPT, students can also gain insights into different cultures and linguistic structures. (iv) Teachers are suggested to incorporate ChatGPT into their curriculum as a supplementary support and to make sure their students are exposed to authentic and real-life communicative language practice. Therefore, it is essential to balance the use of ChatGPT and the development of learner autonomy to prevent the possible deterioration of language skills, higher-level thinking skills, and the over-reliance on ChatGPT.

4.5 Theoretical implications

The integration of ChatGPT into the foreign language learning process presents various theoretical implications for language education. (i) Firstly, ChatGPT appears to be compatible with personalized learning approaches. According to the personalized learning principles highlighted by Walkington and Bernacki (2020), ChatGPT can offer individualized feedback and guidance by providing responses tailored to individual prompts. This can lead to a more meaningful and effective

learning experience. (ii) In line with the Self-Regulated Learning theory developed by Zimmerman and Pons (1986), ChatGPT can guide students in learning strategies and time management. (iii) Within the context of Cognitive Load Theory (CLT) (Sweller, 1988), the adaptable nature of ChatGPT can optimize cognitive processing by ensuring that the information presented is neither too complex nor overly simplified. (iv) Lastly, under the Engagement Theory proposed by Kearsley and Shneiderman (1998), the interactive nature of ChatGPT, by providing prompt feedback and interactive learning experiences, can enhance students' active participation in the learning process.

5 Conclusion

In conclusion, we explored the impact of ChatGPT-assisted foreign language learning experiences through the views of university students on the opportunities and challenges of ChatGPT-assisted language learning. The findings revealed ChatGPT's positive impact on writing and a limited effect on speaking skills, developing grammar and word skills, motivating the participants in online classes, and promoting interactive language learning experiences. Moreover, ChatGPT was found to contribute to the participants' vision of diverse cultures and fostered their creative thinking skills during classroom activities. However, a significant concern that emerged from the results was the potential over-reliance on ChatGPT. This concern aligns with existing literature indicating that over-reliance on AI tools may lead to skill deterioration and hinder the development of critical thinking and learner autonomy. The participants' apprehensions underscore the need for a balanced approach in integrating AI technologies like ChatGPT into language learning curricula. Reflecting on these insights, the conclusion extends to emphasize the importance of educators' role in ensuring that AI tools are used judiciously, complementing rather than replacing the facilitator role of teachers. On the other hand, encountering technical problems in connection and drawbacks in addressing human emotions and personal expressions are additional disadvantages of the tool. To improve ChatGPT-assisted language learning process, suggestions including improving ChatGPT's technical capabilities, accommodating multiple simultaneous users, and incorporating ChatGPT into real-life scenarios for active participation and effective knowledge transfer were made. The limitation of this research is that it was conducted with 13 participants since the necessary permission to conduct the research only allowed the study to be conducted in a single classroom, due to concerns of the administration of the foreign languages department about using ChatGPT in online classes. Despite its limitations, the study contributes to our understanding of the impact of ChatGPT on students' foreign language learning experiences from students' perspectives.

Appendix 1: Semi-structured interview questions

1. Learning a foreign language with ChatGPT:

- What significance does the phrase "Learning a foreign language with ChatGPT" hold for you? Could you elucidate your overall perspective on this matter?

2. The Utilization Domains of ChatGPT:

- For which types of activities and skills do you employ ChatGPT in the realm of foreign language learning?
 - Among these activities, which ones have proven to be most beneficial for you? (For instance, writing practices, conversational exercises, grammar and vocabulary acquisition, etc.)
 - In which linguistic skills has the influence of ChatGPT been most pronounced? Could you expound upon the reasons?

3. The Impacts of ChatGPT on language learning:

- Would you elucidate the impact of ChatGPT on the following aspects?
 - Your foreign language acquisition process.
 - Your proficiency in employing the learned language in real-life scenarios.
 - Your motivation towards learning.

4. The Positive Aspects of ChatGPT in language learning:

- During the foreign language acquisition process, which beneficial and positive aspects of ChatGPT have you observed?
 - Could you share the most poignant moment you've experienced with ChatGPT? What sentiments did that instance evoke within you?

5. The Negative Aspects and Limitations of ChatGPT in language learning:

- In the process of foreign language learning, what negative aspects or limitations of ChatGPT have you observed?
 - Do you believe that the utilization of ChatGPT has led you to adopt habits that could be characterized as negative? Why?

6. More Effective Utilization of ChatGPT in language learning:

- How could ChatGPT be employed more effectively in the foreign language learning process?
 - What would you propose for a more effective utilization of ChatGPT in the language learning process?
 - In your opinion, which enhancements or additions to ChatGPT's features would augment its efficacy in the foreign language learning process?

7. Additional Thoughts and Suggestions in language learning

- Are there any other thoughts, experiences, or suggestions you wish to share pertaining to this subject?

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Data availability The authors confirm that data supporting the findings of this study are available within the article. Raw data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

The authors report no declaration of interest.

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